C-HI 5 Myr Particle-Loss Smoke

Date: 2026-05-10

Summary

This report records the first particle-loss-aware GRAiLE/XRAiSE smoke for the particles-enabled high-power cluster case, `C-HI`, using the same aligned target as the truth, bridge observable, XRAiSE grid, and cavity-protocol smokes:

- grid output: `data.0008.flt.h5`
- grid timestep: `Timestep_8`
- grid time: `4.9994 Myr`
- particle output selected by XRAiSE: `503`
- particle time: `4.9994 Myr`
- view protocol target: `0 deg`
- photon frequency: `2.417990504024e17 Hz`, equivalent to `1 keV`
- redshift: `0.05`

This is not yet the final radio/X-ray map product. It is the runtime gate that proves Georgia's supplied particle-loss machinery can ingest our `C-HI` particle history, run on a `kunanyi` compute node, and produce finite particle-level PRAiSE emissivity values for the same 5 Myr target.

What Was Added

Two small runtime steps were added.

First, the raw PLUTO particle files were converted into the preprocessed HDF5 layout expected by Georgia's driver:

- source particle history: `/scratch/bwbrook/xray_cavities_runs/2026-05-08-c-hi-segment/C-HI/particles/problem/outputs`
- processed particle file: `/scratch/bwbrook/xray_cavities_analysis/2026-05-10-c-hi-particles-hdf5/C-HI.particles.hdf5`
- file size: `129 MB`
- particle IDs represented: `17516`
- particle output times: `550`
- particle time span: `0.0` to `5.4550 Myr`
- fields carried through: `13`

Second, the particle-loss smoke used the processed HDF5 with supplied `xraise.py` and a staged `plutokore` source checkout. The `kunanyi Anaconda3/2024.02-1` module provides the heavy scientific stack. Three tiny compatibility shims were staged for import compatibility only: `contextlib2`, `hdf5plugin`, and `yt`.

Kunanyi Jobs

Particle HDF5 preprocessing:

- job id: `690635.kunanyi-ohpc`
- job name: `c_hi_part_h5`
- node: `n001`
- resources: `select=1:ncpus=1:mpiprocs=1`
- extraction window: `2026-05-10T09:27:15Z` to `2026-05-10T09:28:15Z`
- result: processed particle HDF5 and compact summary archive written

Particle-loss smoke:

- job id: `690636.kunanyi-ohpc`
- job name: `c_hi_ploss5`
- node: `n001`
- resources: `select=1:ncpus=4:mpiprocs=1`
- extraction window: `2026-05-10T09:32:07Z` to `2026-05-10T09:32:35Z`
- result: full-loss particle emissivity array and compact summary archive written

Compact archives:

```
particle HDF5 summary archive:
remote: /u/bwbrook/codex_kunanyi/c_hi_particle_hdf5_smoke_2026-05-10_summary.tar.gz
local:  build/science/kunanyi_stage_2026-05-08/c_hi_particle_hdf5_smoke_690635_summary_2026-0
sha256: 282820847ceb41071cc7199a22eec0ccd5438c91442183095ea09511e24352de

particle-loss archive:
remote: /u/bwbrook/codex_kunanyi/c_hi_xraise_particle_loss_smoke_2026-05-10.tar.gz
local:  build/science/kunanyi_stage_2026-05-08/c_hi_xraise_particle_loss_smoke_690636_2026-05
sha256: be405ff8884c24de2a0e61a2b8cb63c9602106f66bd5bfa47d942abda560c89d
```

The large processed `C-HI.particles.hdf5` is deliberately retained on `kunanyi` scratch and is not copied into the repo.

Scalar Smoke Metrics

Metric	Value
grid output	8
grid time	4.9994358508106655 Myr
particle output	503
particle time	4.999435003568512 Myr
particle emissivity shape	17516 x 1 x 1
finite values	17516
positive values	13144
NaN values	0
minimum	0.0
maximum	4.2874063006224175
mean	0.7759846208637153
positive minimum	0.1851950760834578
positive mean	1.0340951475234963

The output filenames use `loss_factor` because this is a smoke-test lane for the loss-aware component. The array extracted here is the `emis` return from Georgia's `praise(...)` call, converted to SI emissivity units before saving.

Figure

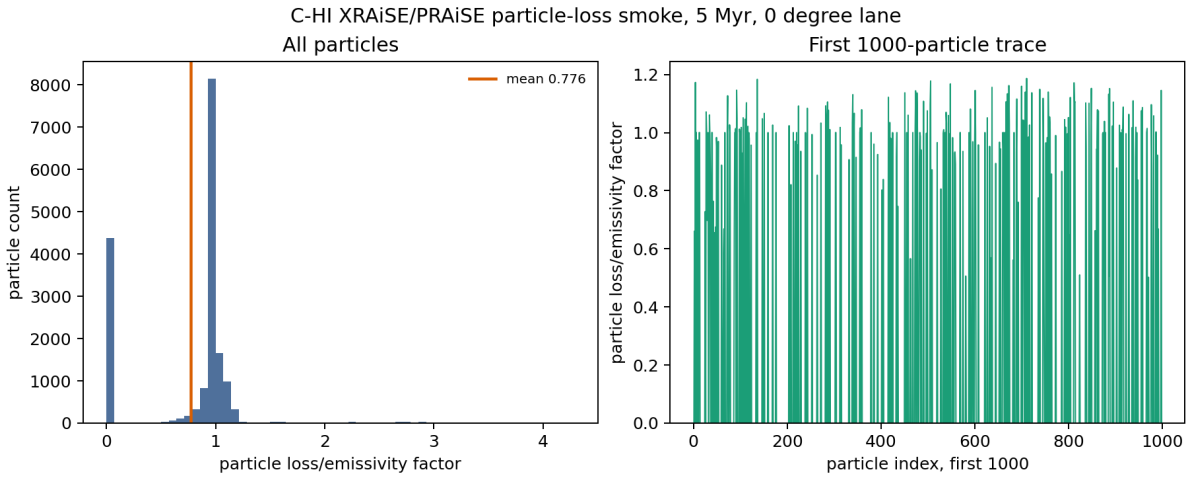


Figure 1: C-HI particle-loss diagnostics

The left panel shows the full finite particle-level distribution. The right panel shows the first 1000 particle entries as a quick visual check that zeros and finite positive entries are both being carried through.

Runtime Warning

The supplied `xraise.py` emitted a `NumbaParallelSafetyWarning` inside `praise_aio(...)` about a variable written inside a parallel loop. The smoke output is finite and reproducible enough for this gate, but before using this path as a production observable pipeline we should run a deterministic cross-check, for example by comparing `NUMBA_NUM_THREADS=1` and `4` on the same snapshot.

Products

The repo-scoped products are stored under:

- `docs/science/assets/c_hi_particle_hdf5_smoke_2026-05-10/`
- `docs/science/assets/c_hi_particle_loss_smoke_2026-05-10/`

Key files:

- `particle_hdf5_summary.csv`
- `particle_hdf5_field_summary.csv`
- `particle_hdf5_metadata.json`
- `particle_loss_smoke_summary.csv`
- `particle_loss_smoke_metadata.json`
- `particle_loss_factor.npy`
- `particle_loss_factor_first1000.tsv`
- `particle_loss_factor_diagnostics.png`

The reusable scripts are:

- `scripts/science/plutokore_particle_hdf5_smoke.py`
- `scripts/science/xraise_particle_loss_smoke.py`
- `scripts/kunanyi/c_hi_particle_hdf5_smoke.pbs`
- `scripts/kunanyi/c_hi_xraise_particle_loss_smoke.pbs`
- `scripts/kunanyi/python_compat/`

Interpretation

This closes the particle-loss runtime gate for the 5 Myr `C-HI` target:

- `plutokore` can be staged on `kunanyi` without modifying the cluster Python installation
- the raw PLUTO particle sequence can be converted into Georgia's expected `*.particles.hdf5` layout
- the supplied XRAiSE/PRAiSE particle-loss code can load the processed particle history and produce finite full-loss particle emissivities at the aligned `5 Myr` snapshot

- the run remains small enough to use as a repeatable smoke test for future longer snapshots

The next observable step is to use these particle emissivities in a mapped or raytraced radio/lobe product and compare it against the already aligned X-ray, cavity, and truth products for the same snapshot/view.